

Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language
Models for Histopathology Image Analysis

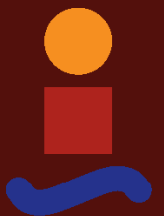
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Dpto. de Ciencias de la Computación e IA
Escuela Técnica Superior de Ingeniería Informática
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Image Analysis

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*Leandro Candau Sánchez de Ybargüen
Sevilla, 2026*

Resumen

- **Contexto.** Clasificación de imágenes histopatológicas bajo el cuello de botella recurrente del dominio: datos etiquetados limitados y vocabularios de etiquetas estrechos.
- **Enfoque.** Aprendizaje contrastivo imagen–texto al estilo CLIP con codificadores fundacionales congelados. Sobre una línea base controlada (ResNet50 + BERT congelados) se evalúa una matriz de ablaciones de una sola variable, entrenadas en Kaggle LC25000 (5 clases de tejido pulmonar y colónico) y evaluadas tanto sobre LC25000 como sobre tres conjuntos externos no vistos en entrenamiento.
- **Resultado principal.** El *preentrenamiento del codificador de imagen* es, con diferencia, la palanca dominante. Manteniendo el resto del pipeline idéntico y sustituyendo el ResNet50 de propósito general (ImageNet) por el ViT-B/32 preentrenado en histopatología de PLIP, la macro-F1 externa mejora en +0.067 en colon de cercanía (NCT-CRC), +0.027 en colon de otro hospital (Chaoyang) y +0.335 en pulmón de otro órgano (LungHist700).
- **Descomposición con DINOv2.** Un ViT-B/14 con preentrenamiento autosupervisado en imágenes naturales generales (sin histopatología) eleva LungHist700 en +0.152 sobre la línea base — aproximadamente la mitad de la ganancia de PLIP. La otra mitad procede del preentrenamiento específico en histopatología.
- **Hallazgos secundarios.** De las otras cinco intervenciones probadas, la normalización de tinción Macenko (referencia NCT-NORM) es la única palanca del lado de la entrada que produce una ganancia positiva medible bajo desplazamiento (+0.030 en Chaoyang, +0.020 en LungHist700), pero su magnitud queda dominada por la palanca del codificador de imagen en los mismos conjuntos y regresa en NCT-CRC (−0.022, que llega ya normalizado por Macenko en su *release*). Los codificadores de texto biomédicos, el clasificador de entropía cruzada sin CLIP y las variaciones de formato de *prompt* no mejoran la generalización. La ventaja de CLIP sobre un clasificador no-CLIP aparece sólo bajo desplazamiento más fuerte, vía la geometría *coseno + L2-normalizada + escalada por temperatura* del clasificador.
- **Aporte metodológico.** El reporte por órgano es esencial: agregar los conjuntos externos en un único número promedio oculta el cambio de régimen entre colon (transferencia parcial con ResNet50/ImageNet congelado) y pulmón (fallo categórico del mismo *backbone*).

Abstract

- **Context.** Histopathology image classification under the domain’s recurring bottleneck: limited labelled data and narrow label vocabularies.
- **Approach.** CLIP-style image–text contrastive learning with frozen foundation-model encoders. A single-variable ablation matrix is evaluated over a controlled baseline (frozen ResNet50 + frozen BERT), trained on Kaggle LC25000 (5 lung and colon tissue classes) and evaluated on both LC25000 and three additional external datasets the model has never seen.
- **Headline result.** The *image encoder’s pretraining* is by far the dominant lever. Holding the rest of the pipeline identical and swapping the general-purpose ResNet50 (ImageNet) for PLIP’s histopathology-pretrained ViT-B/32 lifts external macro-F1 by +0.067 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-organ lung (LungHist700).
- **DINOv2 decomposition.** A ViT-B/14 with self-supervised pretraining on general natural images (no histopathology data) lifts LungHist700 by +0.152 over baseline — roughly half of PLIP’s gain. The other half comes from histopathology-specific pretraining.
- **Secondary findings.** Among the other five interventions tested, Macenko stain normalization (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift (+0.030 on Chaoyang, +0.020 on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (−0.022, which arrives Macenko-normalised at release). Biomedical text encoders, a non-CLIP cross-entropy classifier, and prompt-format variations do not improve generalization. CLIP’s edge over a non-CLIP classifier appears only under heavier shift, via the *bounded-cosine + L2-normalised + temperature-scaled* classifier geometry.
- **Methodological contribution.** Per-organ external reporting is essential: aggregating external datasets into a single mean-OOD number hides the regime change between colon (partial transfer with a frozen ImageNet ResNet50) and lung (categorical failure of the same backbone).

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1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data — a recurring limitation in this domain.
- Foundation models [1] — able to do different tasks with little task-specific supervision; in the vision–language line, CLIP [2] is the paradigm under study here.
- Project question (refined during the project): “*Where does the pretraining of a CLIP-style histopathology pipeline pay off, and where does it not?*” — evaluated across head-side, loss-side, text-side, and image-side interventions, both in-distribution and on three external datasets covering two organs.
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction → semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot → image-level classification, same domain, reframed question.
- Training source: LC25000 [3] (5-class, lung + colon).
- External evaluation: NCT-CRC-HE-7K [4] (close-shift colon), Chaoyang [5] (different-hospital colon), and LungHist700 [6] (different-organ lung).
- Reference points: PLIP [7] zero-shot (no LC25000 supervision) and a plain image classifier (no CLIP architecture).

1.3 Research Questions

- **RQ1** — What baseline classification and generalization behaviour does a frozen general-purpose CLIP pipeline (ImageNet ResNet50 + BERT) achieve on LC25000, and what failure modes appear?
- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learn one task, generalize to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalization help when we evaluate on datasets the model was never trained on?

- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT [8], PubMedBERT [9]), instead of generic BERT [10], improve performance under short class-name prompts?
- **RQ5** — Does the image encoder’s *pretraining choice* matter? Specifically: replacing the general-purpose ResNet50 [11] with PLIP’s histopathology-pretrained ViT-B/32 [7, 12], and with DINOv2’s general-domain self-supervised ViT-B/14 [13] as a control that isolates “ViT architecture and self-supervision” from “histopathology pretraining specifically”.
- **RQ6** — How well does each variant generalize to data the model has never seen, evaluated across three external datasets that span shift severity and two organs?

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment at a time in Chapter 4.

Headline findings (full results in Chapter 4):

- **Image encoder pretraining is the dominant lever.** Replacing ResNet50 with PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) gains +0.067 F1 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-organ lung (LungHist700). No other intervention tested moves the needle comparably.
- **The lung-side advantage decomposes evenly.** The DINOv2 control (same general ViT architecture family as PLIP, self-supervised on general natural images, no histopathology data) lifts LungHist700 by +0.152 over baseline — about half of PLIP’s +0.335 gap. The other half is attributable to histopathology pretraining specifically.
- **Five other interventions move the needle far less, or only in narrow regimes.** Stain Macenko (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift (+0.030 on Chaoyang, +0.020 on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (−0.022, which arrives Macenko-normalised at release). Biomedical text encoders (BioBERT, PubMedBERT) tie or trail baseline across all 6 grid cells on NCT, and the best biomedical cell (PubMedBERT + name_only) extended to Chaoyang and LungHist700 confirms no transfer benefit on any of the three datasets. A non-CLIP plain classifier ties CLIP in-distribution and on close-shift external data but loses on heavier shift; prompt-format variations within the text encoder ablation are within noise.
- **CLIP’s foundational-model premise is validated under shift, not in-distribution.** CLIP and a plain CE classifier tie at saturation in-distribution and on close-shift external; CLIP’s edge appears under heavier shift via the bounded-cosine + temperature-scaled classifier geometry, which doesn’t degenerate the way high-confidence softmax does.

Methodological contributions:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Per-organ external reporting: results are presented per dataset, not averaged; aggregate mean-OOD numbers would hide the cross-organ regime change observed in this study.
- Reproducibility: artifact bundle (fixed-seed split, cached DINOv2 features, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits (Chapter 5).

1.5 Document Structure

- Chapter 1 (Introduction) — motivation, scope, research questions, contributions.
- Chapter 2 (State of the Art) — foundation models, multimodal contrastive lineage, pathology FMs, where this thesis fits.

- Chapter 3 (Methodology) — CLIP architecture, image and text encoders, datasets, preprocessing, baseline, evaluation protocol, reproducibility.
- Chapter 4 (Results and Discussion) — the experiments and what each measured, ending in a synthesis section.
- Chapter 5 (Conclusions and Future Work) — one-paragraph-per-RQ conclusions, limitations, future work.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

BOMMASANI ET AL., 2021

- Foundation models framing (Bommasani et al. [1], 2021).
- CLIP and multimodal contrastive learning (Radford et al. [2], 2021; see also ALIGN [14]) — the paradigm under study.
- Self-supervised visual pretraining — contrastive (SimCLR [15], MoCo [16]), self-distillation (DINO [17]), and scaled SSL on diverse natural images (DINOv2 [13]).
- Pathology shift: per-tile supervised CNNs → foundation models (2016–2024).
- Pathology-specific foundation models:
 - TransPath [18] (2022) — transformer + SSL.
 - Phikon [19] (2023) — masked image modelling.
 - PLIP [7] (2023) — CLIP on medical Twitter.
 - CONCH [20] (2024) — larger pathology VLM.
 - UNI [21] (2024), BiomedCLIP [22] (2023), MUSK [23] (2025) — recent entrants.
- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko [24], Reinhard [25], Vahadane [26]).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
 - Open-vocabulary classification (new classes via prompts proven by adding new samples).
 - Semantic similarity output (cosine vs natural language).
 - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder + shared embedding space.
- InfoNCE loss, learnable temperature.
- In-batch false-negative regime at small batch + few classes.
- Prompt-based zero-shot classification.
- CLIP-paper sample-level retrieval → doesn't apply (class-shared prompts).

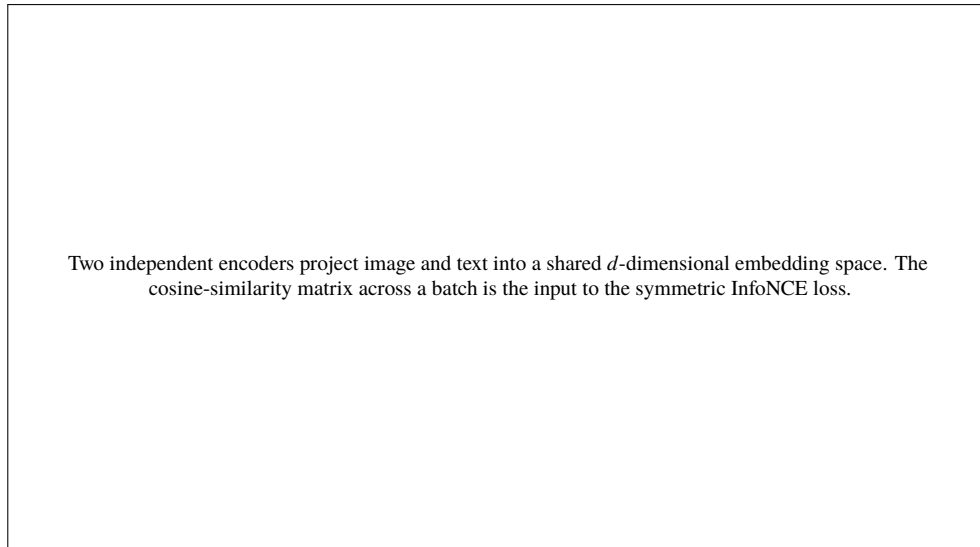


Figure 3.1 CLIP dual-encoder architecture used in this project..

3.3.3 PLIP: a histopathology-pretrained CLIP-style model

- Huang et al. [7] (2023) — a CLIP-style model pretrained on histopathology image–text pairs.
- Training corpus: OpenPath, $\sim 208k$ histopathology image–text pairs scraped from Twitter conversations among pathologists, supplemented with LAION subset filtering.
- Architecture: CLIP ViT-B/32 vision encoder paired with a CLIP-style text encoder, both trained contrastively from a CLIP initialisation.
- Used in this thesis in two distinct ways:
 - As a *zero-shot reference baseline* (§3.6): the full PLIP model (image + text encoders) used off-the-shelf for cosine-similarity classification on all four datasets (LC25000 + the three external evaluation sets).
 - As a *pretrained image encoder* in our image-encoder ablation (§3.4.2): PLIP’s ViT-B/32 vision encoder paired with our trained projection head and a separate BERT text encoder.

3.4 Image Encoders

3.4.1 ResNet50 (CNN, ImageNet supervised)

- He et al. [11] (2016), residual learning.
- $\sim 23M$ parameters, 2048-d output via global average pooling.
- Supervised pretraining on ImageNet-1K ($\sim 1.3M$ natural images, 1000 classes).
- Role in this thesis: general-purpose image-encoder baseline; carries the assumption that ImageNet features transfer broadly.

3.4.2 Vision Transformer (ViT)

- Dosovitskiy et al. [12] (2020).
- Patch-based architecture: image split into $N \times N$ patches, each linearly embedded as a token, then processed by a stack of transformer encoder layers with self-attention.
- The “B” sizing (ViT-Base): 12 layers, hidden size 768.
- The “/16”, “/32”, “/14” suffixes denote patch size; smaller patch = more tokens = finer spatial granularity at the same input resolution.
- Role in this thesis: we use **PLIP’s pretrained ViT-B/32** as one image-encoder variant. PLIP itself is described in §3.3.3; this section covers only the ViT architecture used by PLIP’s vision encoder.

3.4.3 DINOv2 (self-supervised on general natural images)

- Oquab et al. [13] (2024).
- Self-supervised pretraining on LVD-142M (~142M curated general natural images); no labels, no text supervision, no histopathology data.
- Self-distillation between a teacher and student network on augmented crops of the same image.
- We use the base size with patch 14 (**ViT-B/14**).
- Role in this thesis: a control. Same general ViT architecture family as PLIP’s vision encoder, similar self-supervised paradigm, but no histopathology pretraining — isolates “ViT + self-supervision” from “histopathology pretraining specifically” in the image-encoder ablation.

3.5 Text Encoders

- Transformer [27] + BERT [10] — text encoder used by baseline.
- BioBERT [8] / PubMedBERT [9] — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP zero-shot and a plain image classifier

3.6.1 PLIP zero-shot

- Uses the full PLIP model (§3.3.3) — both image and text encoders, frozen — for zero-shot classification.
- Class prediction: cosine similarity between the image embedding and the five PLIP-encoded LC25000 class-prompt embeddings, argmax.
- No LC25000 supervision; no projection head trained on our data.
- Evaluated on all four datasets: LC25000 (training source) plus the three external evaluation datasets (NCT-CRC, Chaoyang, LungHist700; the latter two are queued follow-up runs at the time of writing).
- Role: isolates the value of histopathology pretraining alone, without any of our fine-tuning.

3.6.2 Plain image classifier

- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Architecture: ResNet50 → Global Average Pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Role: isolates what the CLIP paradigm itself contributes once the image encoder, data and adaptation budget are held constant.

3.7 Training Dataset: LC25000

- Borkowski et al. [3] (2019).
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

3.8 External Evaluation Datasets

- Role: data the model has never seen during training. We evaluate every LC25000-trained variant against three external datasets, chosen to span shift severity and two organs.
- Order presented below corresponds to increasing shift severity relative to LC25000.

Five-class sample: "benign lung", "lung adenocarcinoma", "lung squamous cell carcinoma", "benign colon", "colon adenocarcinoma".

Figure 3.2 Example image per class from LC25000..

Table 3.1 LC25000 class counts and stratified split sizes..

Class	Total	Train	Val	Test
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

3.8.1 NCT-CRC-HE-7K

- Kather et al. [4] (2019), Zenodo record 1214456.
- 7 180 patches across 9 tissue classes; we restrict to NORM ($\leftrightarrow$ benign colon tissue) and TUM ($\leftrightarrow$ colon adenocarcinoma) for 1 974 patches scored (741 NORM + 1 233 TUM).
- 224×224 patches at $\sim 20\times$ magnification, Macenko-normalised at release.
- Role: close-shift colon evaluation (stylistically similar acquisition to LC25000).

3.8.2 Chaoyang

- Zhu et al. [5] (2022; preprint 2021).
- 6 160 patches across 4 classes (normal, serrated, adenocarcinoma, adenoma); we restrict to normal ($\leftrightarrow$ benign colon, $n = 1 816$) and adenocarcinoma ($\leftrightarrow$ colon adeno, $n = 2 244$) for 4 060 patches scored.
- 512×512 patches at $20\times$, Beijing Chaoyang Hospital.
- Documented label noise in the training split; the test split is labelled by consensus of three pathologists.
- Role: different-hospital colon evaluation (different scanner, different staining, label noise).

3.8.3 LungHist700

- Tabatabaei et al. [6] (2024, *Scientific Data*).
- ~ 691 high-resolution images at $20\times$ and $40\times$ covering normal lung tissue plus adenocarcinoma and squamous-cell carcinoma at three differentiation grades each; we restrict to the $20\times$ subset for 3-class scoring (359 images: 85 normal + 146 AdC + 128 SqC).
- Role: different-organ lung evaluation; the only multi-class scoring among the external datasets.

3.9 Preprocessing and Input Pipeline

3.9.1 Image preprocessing (per-backbone)

- ResNet50: RGB / 255.0 (i.e., $[0, 1]$ linear scaling).

- ViT-B/32 (PLIP): CLIP-standard mean/std normalisation
mean = [0.481, 0.458, 0.408], std = [0.269, 0.261, 0.276].
- ViT-B/14 (DINOv2): ImageNet mean/std normalisation
mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225].
- All inputs resized to 224×224 bilinear. No additional augmentation (LC25000 already includes rotation/flip augmentation at release).
- Each backbone expects the input distribution it was pretrained on; mismatched preprocessing produces degraded features — see §3.9.4.

3.9.2 Text preprocessing

- All text encoders use BERT-family WordPiece tokenizers (BERT, BioBERT, PubMedBERT, and PLIP's text encoder).
- Fixed sequence length 24 tokens (covers all class-prompt formats).
- Composed prompt format: "A histopathology image of {class_name}."

3.9.3 LC25000-specific cleanup

- 1 280 byte-identical duplicate files identified by MD5 grouping; excluded via `src/data/lc25000_dedupe_exclusions.json`.
- Stratified 80/10/10 split, seed 42, persisted to canonical JSON.
- The canonical split file predates the dedupe pass, so the 1 280 duplicates still cross train/test for the runs reported; flagged as a limitation in Chapter 5.

3.9.4 Methodological note on preprocessing

- Mid-project we caught that the pipeline was applying Caffe-style BGR preprocessing to a backbone expecting [0, 1] RGB — a mismatch from two libraries shipping different ResNet50 checkpoints under the same name.
- The error was surfaced via a linear-probe diagnostic on the frozen features (the probe macro-F1 was well above the trained CLIP ceiling), and fixed with a centralised image loader at `src/data/images.py`.
- Beyond the project-specific bug, the takeaway is that linear-probing a frozen feature extractor against the trained model's ceiling is a cheap sanity check worth running as a project-foundation step.

3.10 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) → 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

3.11 Evaluation Protocol

3.11.1 Metrics and visualizations

- Classification rule (CLIP variants): argmax of cosine similarity between the image embedding and the five LC25000 class-prompt embeddings.
- Classification rule (plain classifier): argmax of softmax over Dense(5).
- Global metrics: accuracy, balanced accuracy, macro-F1, weighted F1.
- Per-class: precision, recall, F1, support.
- Confusion matrices: raw counts + row-normalised by true class.
- Visualizations: shared image + prompt UMAP (training source), and cross-distribution UMAP grid per external dataset (LC25000 dots + external triangles overlaid in one fit).

Table 3.2 Baseline hyperparameters. Ablations change exactly one row..

Hyperparameter	Value
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.11.2 External-dataset evaluation: restricted-argmax protocol

- Each variant exposes all 5 LC25000 class prompts at inference, but on external evaluation we argmax *only* over the LC25000 classes that map to the external dataset’s organ:
 - NCT and Chaoyang (colon datasets): argmax restricted to {benign colon tissue, colon adenocarcinoma}.
 - LungHist700 (lung dataset): argmax restricted to {benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma}.
- Why: external datasets cover only a subset of LC25000 classes; an unrestricted argmax would routinely predict the model’s strongest out-of-organ class on these images and produce uninformative misclassifications.
- Predictions into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis.
- Pure inference: every variant reuses its LC25000-trained checkpoint; no fine-tuning on external sets.
- Class-mapping rules per dataset are detailed in §3.8.

3.11.3 Per-organ reporting

- Results are reported per external dataset, not averaged across datasets.
- Aggregating across datasets would hide the cross-organ regime change observed in this study (Chapter 4); a single mean-OOD number collapses an order-of-magnitude difference between organs into one misleading point.
- Per-dataset reporting is therefore an evaluation choice with explicit methodological motivation.

3.12 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in §4.1.2.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH [20] / BiomedCLIP [22] / MUSK [23]. Deferred to future work.

3.13 Computational Setup and Reproducibility

- Hardware: Google Colab T4 (occasional L4 for longer runs); local CPU/GPU fallback for development and analysis.
- Main stack: TensorFlow + KerasHub for the CLIP-style training pipeline and all downstream evaluation.
- Auxiliary stack: PyTorch + HuggingFace transformers for image-encoder feature extraction when the encoder lacks a TensorFlow port (DINOv2; see below).
- Library versions pinned in `requirements.txt`.
- Reproducibility: seed 42 across NumPy, TensorFlow, and PyTorch; PYTHONHASHSEED set; `tf.config.experimental.enable_op_determinism()`. Runs are bit-exact on the same hardware; small numerical drift can appear across GPU types but does not change the classification outcomes reported.
- DINOv2 has no TensorFlow port in our environment; its image features are precomputed once in PyTorch and cached to disk as HDF5 (one file per dataset, keyed by image relative path). The TensorFlow training and evaluation pipelines then read from cache. Mathematically equivalent to running the frozen backbone online, since the backbone is frozen and each image always produces the same feature vector. Total cache footprint ~ 100 MB across LC25000 plus the three external datasets.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research questions

- **RQ1** — What baseline classification and generalization behavior does a frozen general-purpose CLIP pipeline achieve on LC25000, and what failure modes appear?
- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learn one task, generalize to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalization, an input-side preprocessing that targets a known shortcut in histopathology, help when we evaluate on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT, PubMedBERT), instead of generic BERT, improve performance under short class-name prompts?
- **RQ5** — Does the image encoder’s *pretraining choice* matter? Specifically: (i) replacing the general-purpose ResNet50 with a Visual Transformer pretrained on histopathology image–text pairs (PLIP/OpenPath), and (ii) replacing it with a Visual Transformer pretrained self-supervised on general natural images (DINOv2), as a control that isolates “ViT architecture and self-supervision” from “histopathology pretraining specifically”.
- **RQ6** — How well does each variant generalize to data the model has never seen, evaluated against three external datasets across two organs that span shift severity (close-shift colon, different-hospital colon, different-organ lung)?

4.1.2 Experiment matrix

- Each row below is one experiment. Each one fixes the rest of the pipeline and changes exactly one axis relative to the baseline.
- All variants are evaluated on the same training-source test split (LC25000) and the same three external datasets (Section 4.1.3).

4.1.3 Training source and external datasets

- **Training source:** LC25000 (5 classes, 80/10/10 stratified split, dedupe applied; see Methodology). Macro-F1 on the LC25000 test split is what we call “training source results” below; for the rigor-minded reader this is the in-distribution evaluation.
- **External datasets** (data the model has never seen during training; we call this “generalization to external datasets” below; for the rigor-minded reader these are out-of-distribution evaluations):
 - **NCT-CRC-HE-7K** (close-shift colon, 1 974 patches; mapped 2-class: NORM ↔ benign colon, TUM ↔ colon adenocarcinoma).

Table 4.1 Experiment matrix. Each experiment isolates exactly one axis of variation relative to the baseline. Sections §4.2–§4.7 follow this order, ending with image-encoder pretraining (the chapter’s headline finding)..

Experiment	Image encoder	Text encoder	Loss	Preprocessing	Axis tested
Baseline (§4.2)	ResNet50 (ImageNet)	BERT base	InfoNCE	RGB / 255	reference
Stain normalization (§4.3)	ResNet50 (ImageNet)	BERT base	InfoNCE	Macenko / Reinhard / Vahadane	input-side
PLIP zero-shot (§4.4)	ViT-B/32 (PLIP / OpenPath)	PLIP text encoder	none (zero-shot)	CLIP mean/std	reference: no LC25000 supervision
Foundational vs plain (§4.5)	ResNet50 (ImageNet)	— (no text)	cross-entropy	RGB / 255	architecture: CLIP paradigm vs plain classifier
Text encoder (§4.6)	ResNet50 (ImageNet)	BioBERT / PubMedBERT	InfoNCE	RGB / 255	text encoder pretraining
Image encoder (§4.7)	ViT-B/32 (PLIP); ViT-B/14 (DINOv2)	BERT base	InfoNCE	varies	image encoder pretraining

- **Chaoyang** (different-hospital colon, 4060 patches scored from a 6 160-image release; mapped 2-class: normal ↔ benign colon, adenocarcinoma ↔ colon adeno).
- **LungHist700** (different-organ lung, 359 images at 20× from a ~ 691-image release; mapped 3-class: normal/AdC/SqC ↔ benign lung / lung adeno / lung squamous).
- **Restricted-argmax evaluation:** each variant exposes all 5 LC25000 class prompts at inference; we argmax only over the LC25000 classes that map to the external dataset’s organ (2 colon classes for NCT/Chaoyang, 3 lung classes for LungHist700). Misclassifications into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis.
- **Pure inference:** every variant reuses its LC25000-trained checkpoint. No fine-tuning on the external sets.
- **Per-dataset reporting:** results are presented per dataset, not averaged. Aggregating across datasets would hide the cross-organ regime change surfaced in Section 4.7.

4.1.4 Metrics and visualizations

- **Macro-F1** is the headline metric (class-balanced; reflects per-class precision/recall tradeoffs better than raw accuracy on uneven class supports).
- Accuracy, balanced accuracy, weighted F1 and per-class precision/recall/F1 reported alongside for completeness.
- **Confusion matrices** (raw and row-normalised) per variant per dataset — the most direct view of how a model distributes errors across classes.
- **Training-source UMAP:** image embeddings + class-prompt embeddings projected into the same 2-D fit (cosine metric). Reads: cluster separation, cluster alignment to prompt markers.
- **Cross-distribution UMAP grid** (per OOD dataset, per variant): a 2×3 grid showing LC25000 dots overlaid with external-dataset triangles in one fit. Reads: does the external dataset co-locate with the matching LC25000 class cluster, or sit in its own region of feature space?

4.2 Baseline: frozen ResNet50 and frozen BERT

4.2.1 Configuration

- Frozen ResNet50 image encoder (general-purpose, ImageNet-pretrained).
- Frozen BERT base (uncased) text encoder.
- Trainable projection heads (256-d shared embedding space) and learnable temperature τ .
- Symmetric InfoNCE loss, AdamW optimiser ($\text{lr} = 10^{-3}$, $\text{wd} = 10^{-4}$).

- Max 30 epochs, EarlyStopping (patience=5, restore best weights), seed 42.
- Composed prompt format: “A histopathology image of {class_name}.”.
- Stratified 80/10/10 split.

4.2.2 Training behaviour

- Training and validation InfoNCE loss converge together; the gap is small, no obvious overfitting visible at the converged epoch.
- Learnable temperature τ drifts from its CLIP initialisation of 0.07 toward a smaller converged value, sharpening the cosine-similarity distribution.

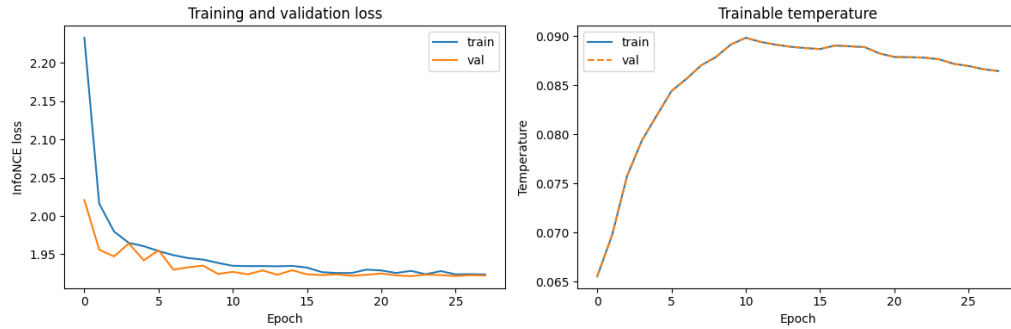


Figure 4.1 Baseline training (blue) and validation (orange) InfoNCE loss across epochs (left), and learnable temperature τ (right). The gap between train and validation is the overfit margin — it remains small here, indicating the projection head + temperature do not over-fit the training set even when the image encoder is frozen..

4.2.3 Training source results

- Macro-F1 on the LC25000 test split: 0.996 (saturated). In-distribution does not discriminate the trained variants in this study — the meaningful comparison is the external evaluation in §??.
- Confusion concentrates on lung_aca (hardest discrimination); benign tissues and lung_scc separate cleanly.

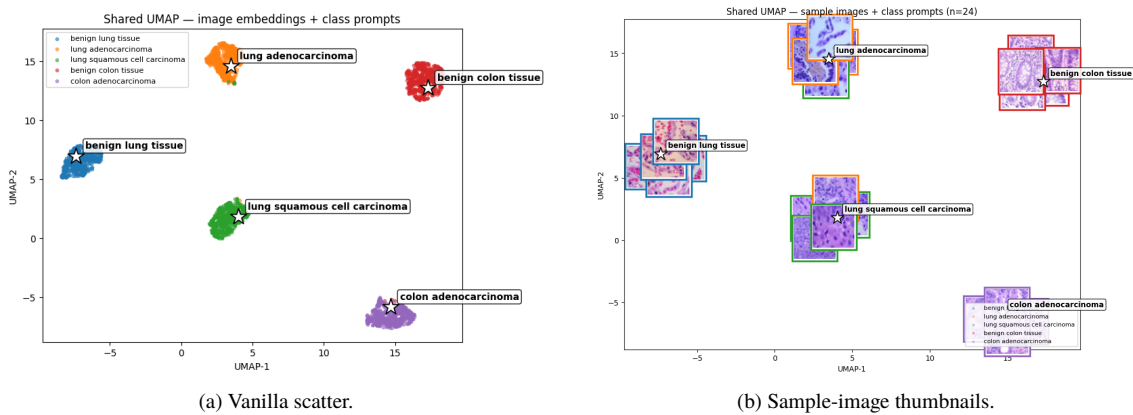


Figure 4.2 Baseline training-source UMAP. Two views from the same fit: (a) per-class scatter of LC25000 test embeddings with class-prompt anchors marked, (b) the same fit overlaid with sample-image thumbnails. The five class clusters are visible but the class-prompt markers collapse to a near-singular point — the short-prompt + frozen text encoder degeneracy that persists across every CLIP-style variant in this chapter..

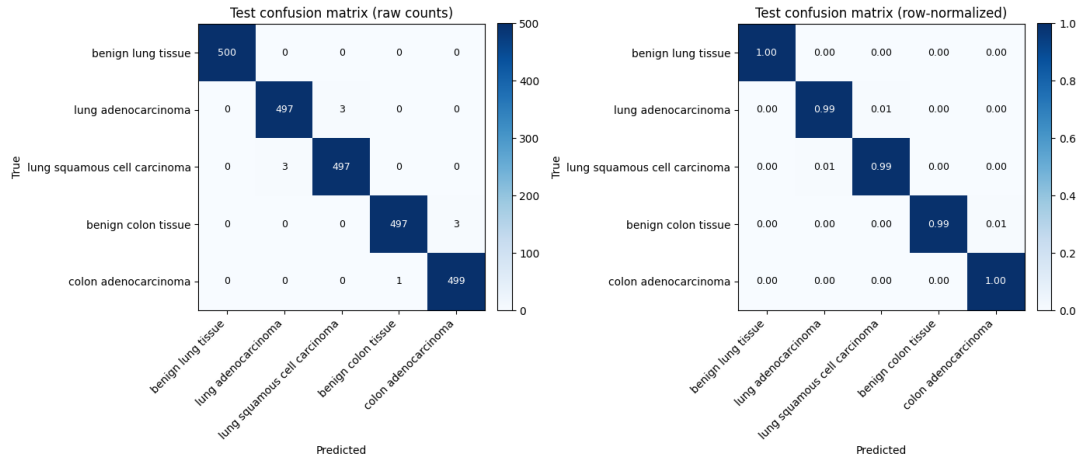


Figure 4.3 Baseline confusion matrix on the LC25000 test split. Left: raw counts. Right: row-normalised by true class. Off-diagonal mass concentrates on lung adenocarcinoma misclassifications..

Table 4.2 Baseline generalization to the three external datasets, restricted-argmax over the target-organ LC25000 classes..

External dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.866	0.870	0.866	0.871
Chaoyang (colon)	0.786	0.796	0.782	0.791
LungHist700 (lung, 20×)	0.305	0.390	0.383	0.301

4.2.4 Generalization to external datasets

- **Colon transfer (NCT + Chaoyang):** macro-F1 in the 0.79–0.87 range. Best on NCT (closest in style to LC25000’s colon distribution); a noticeable drop on Chaoyang despite both being colon (different hospital, different staining, label noise on Chaoyang training split).
- **Lung result (LungHist700):** macro-F1 = 0.305 on a 3-class problem. Per-class recall: SqC 0.89, AdC 0.07, normal 0.19 — the model predicts “squamous” on ~90% of inputs regardless of true class. The baseline does not classify lung tissue on LungHist700; it collapses into a single-class predictor. (The 0.305 figure is barely above the 3-class chance of 0.33; this is a near-chance result.)
- **UMAP signature for the lung collapse** (Figure 4.4): all 359 LungHist700 samples land in a single tight cluster in feature space, with no visible internal class structure. The image encoder produces no usable per-class variance on lung tissue from LungHist700.

4.2.5 What we learned

- Baseline transfers *partially* to colon external data (NCT, Chaoyang), losing ~8 F1 between close-shift (NCT) and different-hospital (Chaoyang).
- Baseline does *not* classify lung tissue on LungHist700 — the model collapses to a degenerate single-class predictor on the lung organ. This is a structural failure of the feature extractor, not a statistical fluctuation.
- Sets up the central question of Section 4.7: is the lung-side failure about the architecture, about the supervised pretraining on natural images, or about the lack of histopathology data in pretraining?

4.3 Stain normalization

4.3.1 Configuration

- Frozen ResNet50 image encoder, frozen BERT text encoder, projection head, symmetric InfoNCE — identical to the baseline (Section 4.2). Only the image inputs change.

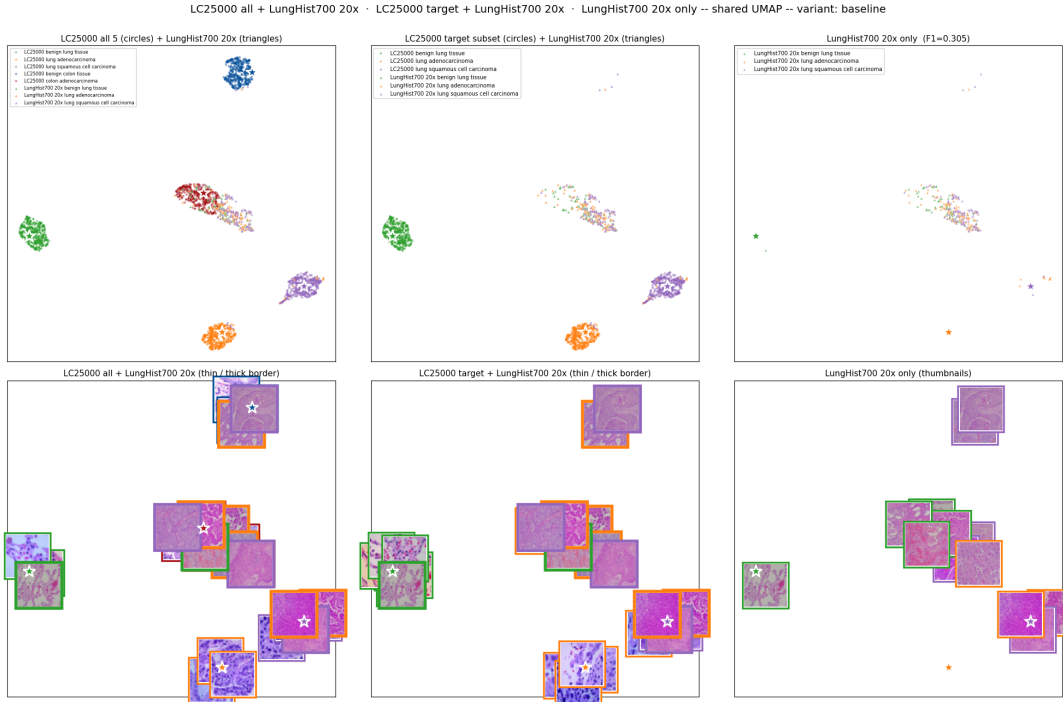


Figure 4.4 Baseline cross-distribution UMAP on LungHist700. LC25000 image embeddings (dots) and LungHist700 image embeddings (triangles) projected into the same 2-D fit. LungHist700 samples cluster in a single tight blob with no visible class structure — the visual signature of a failed feature extractor on this organ..

- **Method:** Macenko normalization, fit on a single reference patch and applied deterministically to every training and evaluation image. The reference is the first NORM tile from NCT-CRC-HE-7K (“NCT-NORM”), chosen so LC25000 lands in approximately the same stain space NCT arrives in — which lets us feed NCT raw at evaluation without a second Macenko pass.
- At external evaluation time, Chaoyang and LungHist700 patches (native stain at release) are passed through the same Macenko(NCT-NORM) transform so they land in the model’s training-stain space. NCT-CRC patches are fed raw (already approximately in that space at release).
- Inputs are cached on Drive after one-time normalization so re-runs of the downstream model are cheap.
- Reinhard and Vahadane were also explored as alternative normalizations in an exploratory pass but did not progress to the full external-evaluation ablation: Vahadane’s NMF solver fails to converge on a fraction of LC25000 patches (visible in Figure 4.6(d)) and Reinhard is a Lab-space colour-statistics transfer rather than a stain-decomposition method, included in the figure for visual comparison only.

4.3.2 Training source results

- Macro-F1 on LC25000 with Macenko(NCT-NORM) at training: 0.986, saturated alongside baseline. In-distribution does not discriminate this lever (or any other trained variant in this study).

4.3.3 Generalization to external datasets

- Three-dataset external evaluation with the Macenko(NCT-NORM) checkpoint, restricted argmax per dataset:
- **The lever’s direction is data-dependent:** it *helps* on the two datasets that arrive in their native stain space (Chaoyang +0.030 F1, LungHist700 +0.020 F1, both receiving Macenko at inference to land in the training-stain space); it *regresses* on NCT-CRC (−0.022 F1), the only dataset that was already partially Macenko-normalised at release.

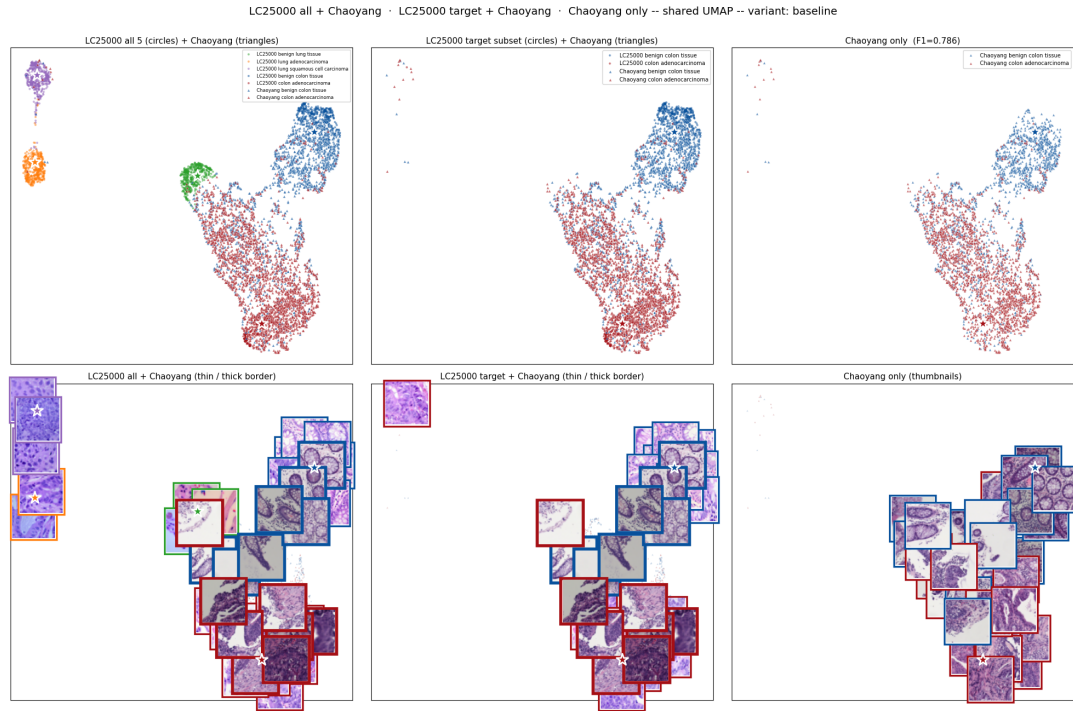


Figure 4.5 Baseline cross-distribution UMAP on Chaoyang. The 2 colon classes are partially separated; Chaoyang triangles sit in their own region adjacent to but not co-located with the matching LC25000 colon clusters..

Table 4.3 Macenko(NCT-NORM) vs baseline on the three external evaluation datasets. Macenko helps where the OOD patches arrive in their native stain palette (Chaoyang, LungHist700) and regresses on NCT, which arrives already Macenko-normalised at release..

External dataset	Baseline F1	Macenko F1	Delta
NCT-CRC-HE-7K (close-shift colon)	0.866	0.844	−0.022
Chaoyang (different-hospital colon)	0.786	0.816	+0.030
LungHist700 (different-organ lung, 20×)	0.305	0.325	+0.020

- **Mechanism:** when the OOD input genuinely sits in a different stain palette than training, Macenko at inference moves it closer to the model’s training distribution and the frozen backbone produces better features. When the input is already in (approximately) the target stain space, applying Macenko at training added a non-linear input-statistic shift that the frozen backbone could not absorb, and the eval-time pass-through cannot recover the small cost paid.
- **Magnitude is modest.** The +0.020–+0.030 F1 gains on Chaoyang and LungHist700 are real but small. By comparison, swapping the image encoder to PLIP (Section 4.7) gains +0.027 on Chaoyang (comparable to Macenko) but +0.335 on LungHist700 (an order of magnitude larger). Stain normalization is not the dominant lever even where it helps.

4.3.4 What we learned

- Stain normalization’s effect under shift is *data-dependent*. It transfers where it normalizes away genuine native-stain differences, and regresses where the input was already in the target stain space (here, NCT-CRC’s release-time Macenko-normalised state).
- Once you commit to Macenko at training time, the right protocol is to apply the same Macenko at evaluation time *only* on OOD inputs that do not arrive in the target stain space already. Doubling normalization on a pre-normalised target is what historically broke the OOD F1 entirely (the “double-Macenko” regime we surfaced earlier in the project).

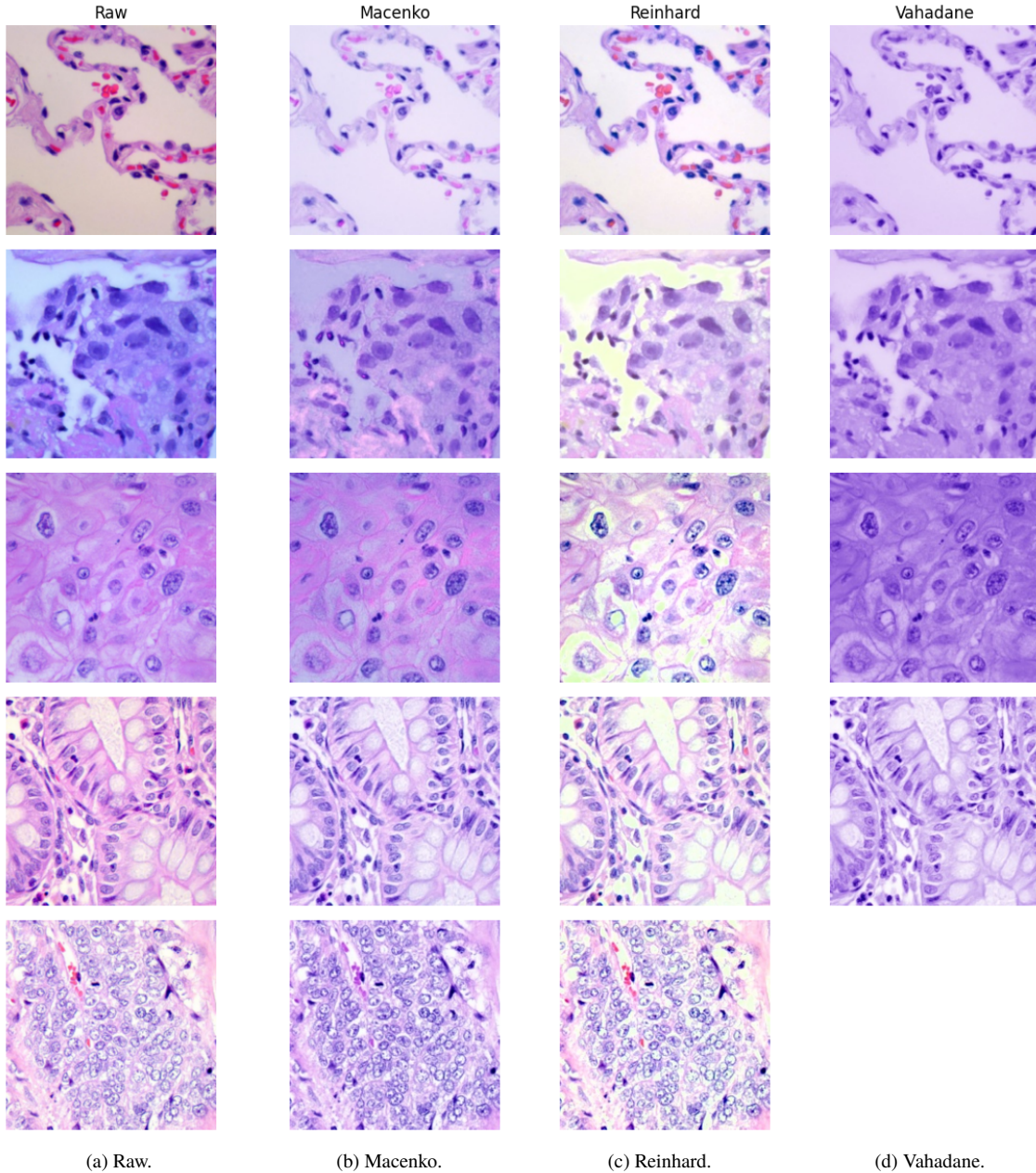


Figure 4.6 Stain normalization, one example per class. Column (a) shows the raw inputs (identical across methods, shown once for compactness); columns (b)–(d) show the same inputs after Macenko, Reinhard, and Vahadane normalization respectively. The bottom-row colon_aca cell in column (d) shows the Vahadane NMF solver failing to converge and re-rendering near-blank, an instability discussed in the interpretation..

- Methodologically: stain normalization is a real, low-cost lever for cross-laboratory histopathology pipelines, but at the frozen-backbone scale evaluated here its gain is dominated by image-encoder pretraining choice on the same datasets.

4.4 PLIP zero-shot reference

4.4.1 Configuration

- Visual Transformer (ViT-B/32) with PLIP’s OpenPath pretraining, paired with PLIP’s matching text encoder. Both frozen.

- Evaluated zero-shot — no LC25000 supervision, no projection head trained on our data. Uses PLIP’s own image and text encoders straight from the model release.
- Class prediction = cosine similarity to the five LC25000 class-prompt embeddings produced by PLIP’s text encoder, argmax.
- Evaluated on the same training-source test split and the same external datasets as every other variant in this chapter.
- **Role in the chapter:** a reference point. PLIP zero-shot isolates the value of pathology pretraining alone — without any of our LC25000 fine-tuning — and shows what an off-the-shelf domain-pretrained model gives you on these tasks.

4.4.2 Training source results

- Macro-F1 on LC25000: **0.700**.
- Pathology pretraining alone (without any LC25000 supervision) clears the saturation null that every *trained* variant reaches at ~ 0.99 by ~ 0.30 F1 in the other direction — but it is the only variant in the chapter where in-distribution numbers are informative, because PLIP zero-shot has no LC25000 supervision to drive the saturation.
- Per-class gains concentrate on benign tissues; cancer subtypes are essentially tied with baseline. The improvement is image-side: PLIP produces visibly better class clusters (Figure 4.7) but the class-prompt embeddings still collapse, same short-prompt degeneracy as the baseline.

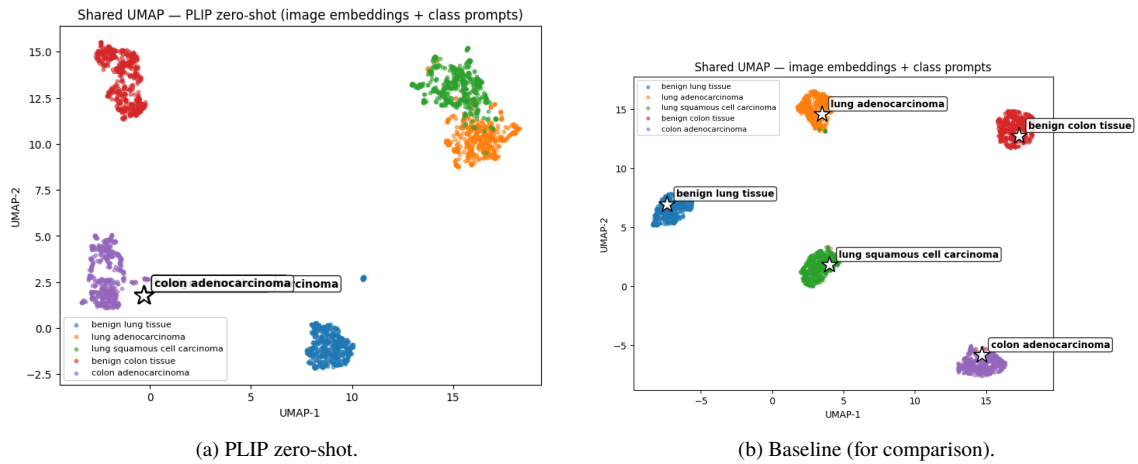


Figure 4.7 Training-source UMAP comparison. PLIP zero-shot (a) shows visibly tighter, more separated class clusters than the baseline (b) without any LC25000 supervision — the gain is image-side..

4.4.3 Generalization to external datasets

Table 4.4 PLIP zero-shot generalization across the three external datasets. No LC25000 supervision; class prediction by cosine similarity to PLIP-encoded prompts..

Dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.661	0.557	0.610	0.679
Chaoyang (colon)	0.731	0.737	0.761	0.727
LungHist700 (lung, 20 $\times$)	0.544	0.621	0.567	0.583

- **Chaoyang > NCT.** PLIP zero-shot scores higher on Chaoyang (0.731) than on NCT-CRC (0.661), opposite of the pattern for every LC25000-trained variant (where NCT is the easier external dataset). Plausible reason: NCT-CRC is Macenko-normalised at release, pushing it away from the natural-staining distribution PLIP saw in OpenPath; Chaoyang’s natural-stain images are stylistically closer to PLIP’s pretraining corpus.

- **LungHist700** = 0.544. The interesting cell. PLIP zero-shot has no LC25000 lung supervision yet outperforms the LC25000-trained ResNet50 variants (0.305) by +0.239 F1 on lung. Per-class recall: benign 0.16 / AdC 0.69 / SqC 0.84 — biased toward squamous like the ResNet50 variants, but not nearly as collapsed.
- Smallest training-source → external drop overall (−0.04 on NCT) — the “pretrained generalist” pattern: never the best, never the worst, but consistent across distributions.

4.4.4 What we learned

- Pathology pretraining produces a *meaningfully better image representation* (visible directly in the UMAP) even without LC25000 supervision.
- PLIP’s off-the-shelf prompt geometry is weak for our specific 5-class LC25000 vocabulary — this decouples “the image representation is useful” from “the off-the-shelf classifier works”.
- Section 4.7 returns to this distinction by keeping PLIP’s image representation but replacing its prompt geometry with a projection head trained on LC25000.

4.5 Foundational model vs plain classifier

4.5.1 Configuration

- Same frozen ResNet50 image encoder as the baseline.
- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Replaces it with a non-CLIP closed-set classifier: ResNet50 → global average pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Isolates what the CLIP paradigm itself contributes once the image encoder, data, and adaptation budget are held constant. Tests RQ2 (the foundational-model premise).

4.5.2 Training source results

- Macro-F1 on LC25000: **0.6369** (vs baseline 0.6273) (*both numbers pre-bug-fix; post-fix both saturate at ~ 0.99 , indistinguishable in-distribution*).
- In-distribution the two architectures tie within noise. The choice between contrastive (CLIP) and direct supervised (plain classifier) is invisible when the image features are fixed and the training distribution is the evaluation distribution.

4.5.3 Generalization to external datasets

Table 4.5 CLIP (baseline) vs plain classifier on the same frozen ResNet50 image encoder, macro-F1 across the training-source test split and three external datasets..

Variant	LC25000	NCT	Chaoyang	LungHist700
Baseline (CLIP InfoNCE)	0.996	0.866	0.786	0.305
Plain classifier (CE)	0.995	0.880	0.722	0.281
Δ (CLIP – plain)	−0.001	−0.014	+0.064	+0.024

- **Training source:** tied at ~ 0.99 (saturation null).
- **NCT-CRC (closest external dataset):** plain CE marginally ahead by +0.014 F1.
- **Chaoyang (harder external):** CLIP ahead by +0.064 F1 — plain CE drops to last.
- **LungHist700 (hardest external):** both variants collapse near 3-class chance; CLIP retains a marginal edge.

Per-class signature on Chaoyang. Plain CE shows benign precision 0.96, benign recall 0.47 (high precision, low recall — the classifier is rarely predicting benign but is right when it does); adenocarcinoma precision 0.70, recall 0.98. The plain classifier is almost-degenerate “predict adenocarcinoma” under shift. The baseline CLIP variant is imperfect but balanced (benign 0.86/0.65, adeno 0.76/0.92), avoiding the collapse.

4.5.4 What we learned

- CLIP and plain CE are *interchangeable on easy distributions*; the choice of architecture is invisible when the model is evaluated on its training distribution.
- CLIP’s edge appears *under shift severity and grows with it*: -0.014 F1 on close-shift NCT (plain ahead), $+0.064$ on different-hospital Chaoyang (CLIP ahead), $+0.024$ on different-organ LungHist700.
- **Mechanism**: same image encoder, same image features. The difference is the classifier geometry. The plain CE softmax learns class *direction vectors* in feature space and is trained to maximise the logit margin; on shifted data those vectors don’t move but the inputs sit in different regions, and the high-confidence softmax collapses into a single-class predictor. CLIP uses cosine similarity to class-prompt embeddings in an L2-normalized 256-d projection space, scaled by a learned temperature. Three effects make this geometry more robust:
 - Cosine similarity is bounded in $[-1, 1]$ — no unbounded margin for the model to over-extrapolate.
 - The L2-normalized space concentrates features near the unit sphere; samples that drift in absolute feature space remain closer in *angular* terms to the prompt anchors.
 - The learned temperature τ acts as a soft regularizer on the cosine distribution: it tempers extreme confidence without removing the ranking signal.
- This is a property of the CLIP *paradigm*, not of any specific encoder. The same geometry was designed for the open-vocabulary regime (new class prompts at inference time) and incidentally protects against distribution shift on closed-set tasks. **This validates RQ2’s foundational-model framing**: CLIP earns its complexity not at peak in-distribution accuracy, but in the open-vocabulary capabilities (retrieval, prompt-tuning at inference) and the robustness-under-shift behaviour shown here.

4.6 Text encoder pretraining

4.6.1 Configuration

- Same arch / loss / prompts / split as the baseline; only the frozen text encoder changes.
- Three encoders: bert-base-uncased (the baseline’s text encoder), BioBERT, PubMedBERT.
- Two prompt strategies: **name_only** (just the class name, e.g. ‘benign lung tissue’) and **composed** (‘A histopathology image of {class}.’, the baseline format).
- $3 \times 2 = 6$ cells. All three encoders are BERT-family, all produce 768-d hidden state, all tokenize through BERT-family WordPiece tokenizers.
- Tests RQ4 (does biomedical text pretraining help under short class-name prompts?).

4.6.2 NCT-CRC grid (6 cells)

Table 4.6 6-cell biomedical text encoder ablation, macro-F1 on NCT-CRC. **None of the 6 cells beats the baseline (0.866 on NCT).** LC25000 in-distribution saturates uniformly; all generalization numbers land in the 0.81–0.86 band. The BERT + composed cell is essentially baseline (same architecture, same prompt template); the best *biomedical* cell — PubMedBERT + name_only at 0.855 — still trails baseline by -0.011 F1..

Cell	LC25000	NCT-CRC	Δ vs baseline NCT
BERT + name_only	0.999	0.821	-0.045
BERT + composed	0.999	0.863	-0.003
BioBERT + name_only	0.998	0.840	-0.026
BioBERT + composed	0.998	0.818	-0.048
PubMedBERT + name_only	0.995	0.855	-0.011
PubMedBERT + composed	0.996	0.808	-0.058

- LC25000 in-distribution: all cells saturate uniformly. The training-source axis does not discriminate these encoders.
- NCT generalization: all 6 cells land in the 0.81–0.86 band. None beats the baseline.

4.6.3 Generalization to Chaoyang and LungHist700

- The best biomedical cell on NCT — PubMedBERT + name_only — was extended to Chaoyang and LungHist700 to test whether biomedical text encoders carry any advantage under heavier shift:

Table 4.7 Best biomedical text-encoder cell (PubMedBERT + name_only) vs baseline on the three external evaluation datasets. The biomedical text encoder ties baseline on Chaoyang (within noise) and regresses on both NCT and LungHist700. No measurable transfer benefit on any of the three datasets..

External dataset	Baseline F1	PubMedBERT+name_only F1	Delta
NCT-CRC-HE-7K (close-shift colon)	0.866	0.855	−0.011
Chaoyang (different-hospital colon)	0.786	0.795	+0.009
LungHist700 (different-organ lung, 20×)	0.305	0.279	−0.026

- **No transfer.** The best biomedical cell ties baseline on Chaoyang (+0.009, within noise) and *regresses* on LungHist700 (−0.026). Combined with the NCT result (−0.011), no biomedical-text-encoder configuration tested produces a measurable gain on any external dataset.
- **LungHist700 collapse signature.** Both baseline and PubMedBERT+name_only collapse on lung, just to different classes. Baseline predicts “lung squamous cell carcinoma” on most inputs in the degenerate regime; PubMedBERT+name_only also collapses to lung_scc but harder ($\sim 92\%$ of predictions). In both cases the per-class recall on the other two lung classes is near-zero — a single-class predictor regime where the text encoder choice only affects how *sharply* the collapse manifests.

4.6.4 What we learned

- Biomedical text pretraining does *not* help on this task at frozen-backbone scale with short class-name prompts. Confirmed across three external datasets: PubMedBERT + name_only (the best biomedical cell on NCT) ties baseline on Chaoyang and regresses on both NCT and LungHist700.
- **Mechanism — why doesn’t “a more domain-aware text encoder” improve results?** Three converging reasons:
 - *The projection head absorbs encoder-specific differences.* BERT, BioBERT, and PubMedBERT all start from different 768-d embedding spaces, but they all get pulled through the same projection head trained on LC25000. The head learns to map any reasonable starting embedding to useful class anchors. With only 5 LC25000 classes to separate, this is an easy adaptation regardless of where the starting embeddings sit.
 - *Short class-name prompts don’t expose biomedical-vocabulary richness.* Biomedical pretraining adds value most when domain vocabulary actually appears — clinical notes, paper abstracts, multi-sentence text. Strings like “colon adenocarcinoma” tokenize into 2–3 wordpieces that any BERT-family encoder handles competently. There is no rare-medical-term upside for biomedical pretraining to surface.
 - *CLIP’s shift robustness is geometric, not domain-knowledge driven.* As argued in Section 4.5, CLIP’s edge under shift comes from the cosine + L2-norm + temperature classifier geometry, which is independent of the text encoder. Swapping in a more medically-aware text encoder doesn’t change the classifier geometry; it changes the starting position of the prompt anchors, which the projection head then re-maps anyway.
- **Small but interesting secondary observation:** vanilla BERT prefers the longer **composed** prompt format; biomedical encoders prefer **bare class names**. Plausible mechanism: biomedical pretraining made strings like “colon adenocarcinoma” carry strong signal on their own, while adding generic boilerplate (“A histopathology image of ...”) dilutes the prompt embedding for biomedical encoders. Not enough to flip any cell to winning — but a useful methodological note for short-prompt design.

4.7 Image encoder pretraining

4.7.1 Configuration

- Same arch / loss / prompts / split / text encoder (frozen BERT base) as the baseline. The only axis that changes is the image encoder’s pretraining.
- Three image encoders compared, all frozen, trainable parameters limited to the projection head + temperature:
 - **ResNet50 (general-purpose, ImageNet pretraining)** — the baseline.
 - **Visual Transformer (ViT-B/32) with PLIP’s OpenPath pretraining** — a ViT pretrained on ~208k histopathology image–text pairs scraped from Twitter.
 - **Visual Transformer (ViT-B/14) with DINOv2 self-supervised pretraining** — a ViT pretrained on ~142M curated general natural images via self-distillation, no labels, no histopathology data.
- **Role of DINOv2 in the comparison:** a control. PLIP differs from the baseline on multiple axes simultaneously (architecture: ViT vs CNN; objective: contrastive vs supervised; pretraining domain: histopathology vs natural images). DINOv2 isolates the “ViT + self-supervised on general data” axis from the “histopathology pretraining specifically” axis. If DINOv2 also breaks the baseline’s lung collapse, the win is partly architectural; if it does not, the win is specifically about histopathology pretraining.
- **Caveat:** the two ViTs differ in patch size (PLIP B/32, DINOv2 B/14). At 224×224 input this gives different feature granularity (49 vs 256 patch tokens). We discuss this as a confound in Methodology and address it as a limitation in Chapter 5.

4.7.2 Training source results

- PLIP image encoder + LC25000 supervision: LC25000 macro-F1 = 0.996. Same saturation null as every other trained variant.
- DINOv2 image encoder + LC25000 supervision: LC25000 macro-F1 = 0.989. The saturation null holds even though DINOv2 has never seen labelled histology — its pretraining is self-supervised on general natural images, with no class labels. Confirms that the in-distribution null is backbone-independent: the bottleneck on the meaningful axis is not what the image encoder “knows about histology” in the supervised sense; it is what kind of variation the encoder learned to be sensitive to.

4.7.3 Generalization to external datasets

Table 4.8 Image encoder pretraining swap, macro-F1 across the training-source split and three external datasets. Everything except the image encoder is identical across rows..

Image encoder	LC25000	NCT	Chaoyang	LungHist700
ResNet50 (ImageNet)	0.996	0.866	0.786	0.305
ViT-B/14 (DINOv2 / self-supervised)	0.989	0.855	0.776	0.457
ViT-B/32 (PLIP / OpenPath)	0.996	0.933	0.813	0.640

- **NCT-CRC (close-shift colon):** PLIP wins by +0.067 F1 over the ResNet50 baseline. DINOv2 lands at -0.011 vs baseline — general-domain self-supervision does *not* help on the easiest external dataset. The colon ceiling around 0.88 that none of the other interventions could clear is broken specifically by changing the image encoder’s pretraining *domain*, not by architecture or self-supervision alone.
- **Chaoyang (different-hospital colon):** PLIP wins by +0.027 F1; DINOv2 lands at -0.010 vs baseline. Same colon-side pattern as NCT: only histopathology pretraining helps; general SSL is interchangeable with ImageNet supervision.
- **LungHist700 (different-organ lung):** this is where the control pays off. PLIP wins by +0.335 F1 over baseline; DINOv2 wins by +0.152 F1. *Roughly half of PLIP’s lung advantage comes from architecture + self-supervised pretraining on general images; the other half comes from histopathology pretraining specifically.* DINOv2 partially rescues the lung collapse (per-class recall 0.46/0.52/0.40, balanced) but does not match PLIP’s full functional classifier (0.51/0.49/0.91).

Per-class signature on LungHist700. Both ResNet50 variants converge to the same degenerate predictor on lung tissue: baseline recall 0.19/0.07/0.89 for normal/AdC/SqC, plain CE 0.15/0.05/0.91 — both predict “squamous” on $\sim 90\%$ of inputs regardless of true class. **DINOv2-encoded variant:** 0.46/0.52/0.40 — balanced across all 3 classes, no degenerate collapse, but lower peak per-class scores than PLIP. **PLIP-encoded variant:** 0.51/0.49/0.91 — balanced *and* retains the high squamous recall, the only variant that does both.

4.7.4 What we learned

- The **image encoder’s pretraining is the dominant lever in this study**, but the DINOv2 control reveals that “pretraining” decomposes into two contributing components.
- Effect-size summary: PLIP gains over baseline by +0.067 F1 on close-shift colon, +0.027 on different-hospital colon, +0.335 **on different-organ lung**. DINOv2 gains over baseline by $-0.011 / -0.010 / +0.152$ on the same three datasets.
- **On colon (NCT, Chaoyang):** only histopathology pretraining moves the needle. General-domain self-supervised pretraining (DINOv2) lands within noise of ResNet50 supervised pretraining on natural images. The colon advantage is specifically about *what* the encoder was pretrained on, not *how*.
- **On lung (LungHist700):** the DINOv2 and PLIP-zero-shot controls together yield a clean three-step decomposition of PLIP-backbone’s +0.335 F1 gap. Baseline at 0.305 $\rightarrow$ DINOv2 at 0.457 (+0.152, “ViT + general-domain SSL” contribution) $\rightarrow$ PLIP zero-shot at 0.544 (+0.087, “histopathology pre-training without LC25000 supervision” contribution) $\rightarrow$ PLIP-backbone at 0.640 (+0.096, “LC25000 supervision on top of histopathology pretraining” contribution). Three roughly comparable steps of ~ 0.1 F1 each, sequentially adding architecture/SSL, then domain, then task supervision.
- Why colon and lung behave differently, on the ResNet50 baseline: ImageNet supervision contains no histology. LC25000 supervision rescues ResNet50 features that happen to correlate with colon class labels in the LC25000 training distribution. That correlation transfers partially to NCT/Chaoyang (both colon, stylistically close). It does not transfer to lung tissue at all — the lung-side “success” in-distribution is memorisation on the LC25000 lung subset, not learned lung tissue features.
- Why DINOv2 helps on lung but not colon: DINOv2’s self-supervised pretraining on $\sim 142\text{M}$ diverse natural images produces image features that are sensitive to a much broader range of local-texture variation than supervised ImageNet pretraining. That broader feature space provides *enough* signal for LC25000 supervision + the projection head to construct a usable lung classifier, even though the encoder has never seen histology. On colon, where ResNet50 features were already enough for partial transfer, DINOv2’s broader features do not help further — the bottleneck on colon is the pretraining domain, not the architecture’s feature breadth.
- Patch-size confound check: DINOv2 has more spatial detail than PLIP (256 vs 49 patch tokens at the same 224×224 input), but still loses on colon. The patch-size difference is therefore not the dominant factor here — pretraining is.

4.8 Discussion

4.8.1 Synthesis across experiments

- This chapter evaluated six interventions on a frozen general-purpose CLIP pipeline (ResNet50 + BERT). Five of the six (stain normalization, biomedical text encoders, plain-classifier substitution, the prompt-format variations within the text-encoder ablation) produce changes within ~ 0.03 F1 of the baseline on close-shift external evaluation (NCT) and at most ~ 0.03 F1 of improvement on the harder external datasets. Stain Macenko is the only input-side lever that moves the needle positively under shift (+0.030 on Chaoyang, +0.020 on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets. The best biomedical text-encoder cell (PubMedBERT + name_only) confirms the null result across all 3 external datasets — ties baseline on Chaoyang, regresses on NCT and LungHist700.
- The sixth intervention — replacing the image encoder’s pretraining (Section 4.7) — moves the headline metric by +0.067 on close-shift colon (NCT), +0.027 on different-hospital colon (Chaoyang), and

+0.335 **on different-organ lung (LungHist700)**, taking the lung result from a degenerate single-class predictor at 0.305 to a functional 3-class classifier at 0.640.

- The DINOv2 and PLIP-zero-shot controls together decompose that gain on lung into a clean ladder: 0.305 (baseline) $\rightarrow$ 0.457 (DINOv2: ViT + SSL contribution, +0.152) $\rightarrow$ 0.544 (PLIP zero-shot: histopath pretraining contribution, +0.087) $\rightarrow$ 0.640 (PLIP-backbone: LC25000 supervision on top, +0.096). Three roughly equal steps of ~ 0.1 F1. On colon, the decomposition collapses — DINOv2 ties baseline and PLIP zero-shot actually falls below baseline on Chaoyang (0.731 vs 0.786) because LC25000 supervision was already enough.
- **The image encoder’s pretraining is the dominant lever**, by roughly $5\times$ over the next-largest single lever on colon external evaluation. On lung, no head/loss/text-side intervention rescues a non-functional predictor at all; only changing the encoder does.
- The contemporary methods literature’s emphasis on novel loss functions, text-side innovations, and stain harmonisation is not supported by this study at the frozen-backbone scale we evaluated.

4.8.2 Secondary observations

- **Cross-organ regime change.** ResNet50 variants partially work on colon external data (0.78–0.88) but *categorically fail* on lung (~ 0.30 , near 3-class chance). Mechanism: LC25000 supervision can rescue ResNet50 features that happen to correlate with colon class labels in LC25000, but cannot produce useful lung-tissue features within a frozen ImageNet ResNet50; the in-distribution lung “success” is memorisation. Per-organ external evaluation is therefore essential; aggregating across datasets hides the regime change.
- **Classification quality $\neq$ representation alignment.** Baseline CLIP achieves cross-dataset *cluster alignment* but lower classification F1; PLIP achieves the highest F1 but leaves external samples in their own region of feature space. The contrastive objective optimises same-class-pair alignment; the cosine-classifier rule only needs alignment to the class-prompt direction. These are different geometric properties — the right tradeoff depends on whether the downstream task is classification or representation-driven (retrieval, clustering, transfer).
- **Plain-classifier flip under shift.** Plain CE ties CLIP in-distribution and marginally wins NCT, but loses Chaoyang (+0.064 for CLIP) and LungHist700 (+0.024). High-confidence softmax classifiers degenerate into single-class predictors as shift severity grows (visible in the Chaoyang per-class signature); CLIP’s cosine geometry doesn’t. This supports the foundational-model framing of RQ2: CLIP earns its complexity in shift robustness, not in-distribution accuracy.

4.8.3 Headline summary

Table 4.9 Headline summary across the chapter. PLIP-pretrained ViT wins every external evaluation; the lung-side advantage (+0.335 F1 over baseline) is an order of magnitude larger than the colon-side advantages. DINOv2 (the same-architecture, general-domain self-supervised control) lands between baseline and PLIP on lung but ties baseline on colon, decomposing PLIP’s lung advantage into roughly equal architecture/SSL and histopathology-pretraining contributions. Stain Macenko (NCT-NORM) helps on the two datasets that arrive in their native stain palette (Chaoyang +0.030, LungHist700 +0.020) and regresses on NCT (−0.022), which arrives already Macenko-normalised at release. The best biomedical text-encoder cell (PubMedBERT + name_only) ties baseline on Chaoyang and regresses on both NCT and LungHist700 — no measurable transfer benefit on any of the three datasets. Limitations and future work are discussed in Chapter 5..

Variant	LC25000	NCT	Chaoyang	LungHist700
ViT-B/32 (PLIP)	0.996	0.933	0.813	0.640
ViT-B/14 (DINOv2)	0.989	0.855	0.776	0.457
Baseline (ResNet50 + BERT)	0.996	0.866	0.786	0.305
Plain CE classifier	0.995	0.880	0.722	0.281
Stain Macenko (NCT-NORM reference)	0.986	0.844	0.816	0.325
Best biomedical text-encoder cell (PubMedBERT + name_only)	0.995	0.855	0.795	0.279
PLIP zero-shot (no LC25000 supervision)	0.700	0.661	0.731	0.544



Figure 4.8 6-cell UMAP grid (one panel per BERT/BioBERT/PubMedBERT $\times$ name_only/composed cell), cross-distribution view on NCT-CRC. The clusters are qualitatively similar across cells, consistent with the lack of meaningful F1 separation in Table 4.6..

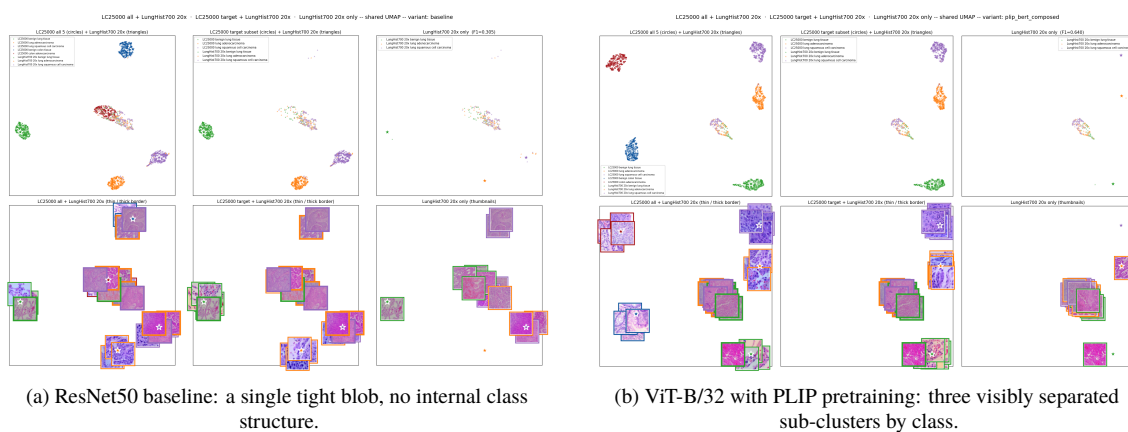


Figure 4.9 LungHist700 cross-distribution UMAP comparison — the visual headline of this chapter. The ResNet50 baseline (a) places all 359 LungHist700 samples into a single tight cluster with no internal class structure; the PLIP-pretrained ViT (b) produces three visibly distinct sub-clusters by class, sitting adjacent to (but not co-located with) the matching LC25000 lung clusters..

5 Conclusions and Future Work

5.1 Conclusions

Headline. The image encoder’s pretraining is the dominant lever in this study. Holding everything else fixed, swapping the general-purpose ResNet50 (ImageNet supervised) for PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) improves external evaluation by +0.067 F1 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-organ lung (LungHist700). None of the other five interventions tested moves the needle comparably. The DINOv2 control reveals that roughly half of PLIP’s lung advantage comes from architecture + self-supervised pretraining alone, the other half from histopathology pretraining specifically.

RQ1 — Baseline behavior and failure modes

- The frozen general-purpose CLIP pipeline (ResNet50 + BERT) saturates LC25000 at ~ 0.99 macro-F1, transfers partially to colon external data (0.866 NCT, 0.786 Chaoyang), and categorically fails on lung external data (0.305 on LungHist700, near 3-class chance).
- The lung failure is a regime change, not a smooth degradation: both ResNet50-based variants collapse into a single-class predictor (predicting “squamous” on $\sim 90\%$ of inputs regardless of true class).
- See Results §4.2.

RQ2 — CLIP’s foundational-model premise

- Validated under shift, not in-distribution. CLIP and a plain CE classifier on the same frozen backbone tie at saturation in-distribution and tie marginally on close-shift external (NCT, plain CE +0.014); CLIP’s edge appears under heavier shift (Chaoyang +0.064 for CLIP, LungHist700 +0.024 for CLIP).
- Mechanism: CLIP’s bounded-cosine + L2-normalised + temperature-scaled classifier geometry is more robust than the high-confidence softmax, which degenerates under shift into a single-class predictor.
- See Results §4.5.

RQ3 — Stain normalization and generalization

- Direction is data-dependent. Macenko(NCT-NORM) at training, with the same transform applied at inference only on OOD datasets that arrive in their native stain palette, *helps* on Chaoyang (+0.030 F1) and LungHist700 (+0.020 F1) and *regresses* on NCT-CRC (−0.022 F1, which is already Macenko-normalised at release).
- Mechanism: when an OOD input genuinely sits in a different stain palette than the training distribution, applying Macenko at inference moves it closer to the model’s training-stain space and the frozen backbone produces better features. On NCT, where the input was already approximately in the target stain space, training-time Macenko added a non-linear input-statistic shift the frozen backbone could not absorb for no gain at evaluation.

- Magnitude is modest. Even where Macenko helps, the gain is comparable to the image-encoder lever on Chaoyang (+0.027 for PLIP) and an order of magnitude smaller on LungHist700 (+0.335 for PLIP). Stain normalization is a useful but not dominant lever.
- See Results §4.3.

RQ4 — Biomedical text encoder pretraining

- No effect on the meaningful axis. None of the 6 cells (BERT / BioBERT / PubMedBERT $\times$ bare-name / composed-prompt) beats the baseline on NCT external evaluation; all cells fall within 0.81–0.86 macro-F1.
- Extended to 3 external datasets via the best biomedical cell (PubMedBERT + name_only): no transfer benefit. Ties baseline on Chaoyang (+0.009, within noise), regresses on NCT (−0.011) and LungHist700 (−0.026). On LungHist700 the model collapses to a single-class predictor ($\sim 92\%$ squamous), same regime as baseline but with sharper collapse.
- Mechanism: the trained projection head absorbs encoder-specific differences; short class-name prompts don’t expose biomedical vocabulary richness. CLIP’s shift robustness is a property of the classifier geometry, not of the text encoder’s domain knowledge.
- Secondary observation: BERT prefers the longer composed prompt; biomedical encoders prefer bare class names — small but useful for short-prompt design.
- See Results §4.6.

RQ5 — Image encoder pretraining (the dominant lever)

- Replacing ResNet50 with PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) gains +0.067 F1 on NCT, +0.027 on Chaoyang, and +0.335 on LungHist700. None of the head/loss/text-side interventions tested gets close.
- The DINOv2 control (ViT-B/14 with general-domain self-supervised pretraining, no histopathology data) decomposes the lung gain: DINOv2 alone lifts LungHist700 by +0.152 over baseline, indicating that $\sim$ half of PLIP’s lung advantage comes from “ViT + self-supervised pretraining on diverse natural images” and $\sim$ half from “histopathology pretraining specifically”.
- On colon, only histopathology pretraining helps; general-domain SSL ties baseline within noise. The lever’s contribution is dose-dependent on shift severity.
- See Results §4.7.

RQ6 — Generalization across 3 external datasets and 2 organs

- PLIP’s ViT-B/32 wins every external evaluation in the study (0.933 NCT, 0.813 Chaoyang, 0.640 LungHist700). The lung-side advantage over baseline (+0.335 F1) is an order of magnitude larger than the colon-side advantages (+0.067 NCT, +0.027 Chaoyang).
- The cross-organ asymmetry is a regime change: ResNet50-based variants transfer partially to colon (0.78–0.88 F1) but categorically fail on lung (~ 0.30 , near chance). PLIP shows no such asymmetry.
- Methodological consequence: per-organ external reporting is essential; aggregating across datasets would have hidden the regime change.
- See Results §4.7 and §4.8.

5.2 Limitations

- LC25000 is single-institution; no patient-level grouping or clinical validation. Three external datasets (NCT-CRC, Chaoyang, LungHist700) span shift severity and two organs but each is itself a single-institution release.
- LC25000 augmentation-pair leakage: 1 250 source images $\times$ 20 augmentations are scattered randomly across the 80/10/10 split. Likely the dominant inflator of in-distribution numbers; external evaluations are unaffected.

- 1 280 byte-identical LC25000 duplicates cross train/test under the canonical pre-dedupe split. Effect $\sim 1\text{--}2$ F1 points on in-distribution numbers; external evaluations unaffected.
- All variants use a frozen image backbone with a trained projection head and temperature. Full end-to-end fine-tuning is out of scope by design; this study tests the frozen-backbone transfer regime, which is the practical setting for histopathology foundation-model downstream use.
- Single seed per condition; seed variance not estimated. Especially load-bearing for the LungHist700 result ($N = 359$ images, per-class supports 85/146/128). Bootstrap confidence intervals on the headline lung gap are listed as a high-priority future-work item.
- External evaluation uses restricted argmax over the LC25000-mappable classes: 2-class colon for NCT and Chaoyang, 3-class lung for LungHist700. Richer multi-class external scenarios are out of scope.
- Patch-size confound in the image-encoder ablation: PLIP (ViT-B/32) and DINOv2 (ViT-B/14) differ in both pretraining domain and patch size (49 vs 256 tokens at 224×224 input). DINOv2’s finer patches do not rescue the colon-side gap, indicating patch size is not the dominant factor, but a fully controlled comparison would match patch sizes.
- Stain normalization is evaluated with a single NCT-NORM reference patch for the external evaluation; reference rotation could shift the result.
- LungHist700 source bias is not characterised; the lung-side ResNet50 failure could be partly stain-shift artefact in addition to the architecture limitation.
- No human-expert evaluation; no clinical validation.
- No head-to-head against other histopathology foundation models (UNI, CONCH, Virchow-2, BiomedCLIP, MUSK, PathChat) — compute and time scope.

5.3 Future Work

Note: this section will be expanded once the rest of the thesis is locked, to ensure the planned follow-ups reflect the final picture of findings.

Medium-term

- Bootstrap confidence intervals on the headline LungHist700 $+0.335$ F1 gap. ~ 1 hour analysis on saved predictions, no retraining. Defends the claim against the small- N critique ($N = 359$ images, per-class supports 85/146/128).
- Extend the PLIP zero-shot reference to Chaoyang and LungHist700. Closes the gap so the reference is comparable across all four datasets.
- Larger effective batch (gradient accumulation or larger GPU) to test whether the InfoNCE false-negative regime is binding in our setup.
- Prompt-tuning methods (CoOp [28], CoCoOp) under frozen encoders — can a learned prompt close some of the colon-side gap without changing the backbone?

Longer-term

- Architecture-controlled comparison against other histopathology foundation models: UNI [21], CONCH [20], BiomedCLIP [22], MUSK [23]. Disambiguates “PLIP specifically wins” from “any histopathology-pretrained backbone wins”.
- Multi-institution training and evaluation.
- Returning to segmentation with pathology foundation-model backbones.
- CLIP-with-short-labels applied to other low-label scientific domains (e.g., the tutor’s gravitational-physics work).

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4.9	Headline summary across the chapter. PLIP-pretrained ViT wins every external evaluation; the lung-side advantage ($+0.335$ F1 over baseline) is an order of magnitude larger than the colon-side advantages. DINOv2 (the same-architecture, general-domain self-supervised control) lands between baseline and PLIP on lung but ties baseline on colon, decomposing PLIP’s lung advantage into roughly equal architecture/SSL and histopathology-pretraining contributions. Stain Macenko (NCT-NORM) helps on the two datasets that arrive in their native stain palette (Chaoyang $+0.030$, LungHist700 $+0.020$) and regresses on NCT (-0.022), which arrives already Macenko-normalised at release. The best biomedical text-encoder cell (PubMedBERT + name_only) ties baseline on Chaoyang and regresses on both NCT and LungHist700 — no measurable transfer benefit on any of the three datasets. Limitations and future work are discussed in Chapter 5.	28

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